



COMPETITION PACK

Cloud Computing Day 4

2026 GLOBAL SKILLS CHALLENGE

WOLLONGONG, AUSTRALIA
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CONTENTS

CONTENTS	2
PROJECT INFORMATION	3
RULES & REGULATIONS	6
EXPERT CONTACT	ERROR! BOOKMARK NOT DEFINED.
APPENDIX	ERROR! BOOKMARK NOT DEFINED.

PROJECT INFORMATION

Based on the provided marking scheme, plan a 5-member group project that mirrors the competition structure. To ensure equal distribution, each member will take responsibility for one of the primary criteria (A through E), while Criterion F (Troubleshooting/Database Operations) is shared among the group to balance the workload.

This project requires deploying a highly available, secure, and auto-scaling web infrastructure in AWS using **Bash scripts**.

Proposed Project: Automated High-Availability Web Platform

Each student can be assigned a specific role corresponding to a marking criterion. Every role must use **AWS CLI commands within Bash scripts** to accomplish their tasks.

Member	Focus Area	Key Tasks (Bash Scripting)	Weight
Member 1	Networking (A)	Script VPC creation, subnets, IGW, NAT, and route tables.	6.0
Member 2	Deploy App (B)	Script web server installation, RDS creation, and DB connections.	6.0
Member 3	Scalability (C)	Script AMI creation, Target Group, ALB, Launch Template, and ASG.	6.0
Member 4	Security (D)	Script Security Group rules for web server and RDS (least privilege).	6.0
Member 5	Monitoring/Load (E)	Script load test triggers and verify ASG scaling behaviours.	6.0

Note: Criterion F (Database Operations/Troubleshooting) is a team effort where each member ensures their component integrates with the database (e.g., Member 2 connects the app, others verify records).

Implementation Guidelines for the Team

To ensure all members meet the requirements and achieve maximum marks:

- **Standardization:** Create a `main.sh` script that calls each member's individual scripts in the correct order ($A \rightarrow D \rightarrow B \rightarrow C \rightarrow E$).
- **Networking (Member 1):** Must explicitly script the CIDR assignments and verify the region is set to `us-east-1`.
- **Security (Member 4):** ~~Scripts must avoid "wildcard" rules (e.g., 0.0.0.0/0 on SSH) and~~ instead implement specific port access (3306 for RDS) as per the criteria.
- **Automation (Members 3 & 5):** The ASG must be configured so that it reacts to load tests; Member 5's script must automate the "navigate out of load test" and verify instance termination.
- **Verification (All):** Every script should include "verification" logic - using `aws cli` commands to query the status of resources and output "Success" or "Failure" to the console.

Managing the Workflow

To keep the group on track, focus on these milestones:

1. **Phase 1 (Setup):** Member 1 provisions the foundation.
2. **Phase 2 (Hardening):** Member 4 applies security policies to the VPC and subnets.
3. **Phase 3 (Logic):** Member 2 and 3 deploy the application and the scaling infrastructure.
4. **Phase 4 (Testing):** Member 5 runs the load test, and the whole team audits the system against the "Judgment" criteria (e.g., "Confirm extra instances are created").

By breaking the project into these five distinct, script-heavy roles, each member manages a comparable volume of code, satisfies the requirement for AWS Bash automation, and is accountable for exactly 6 marks toward the total 30-mark pool.

Tips for the Team

- **Modularization:** Have each member save their scripts in a central repository (e.g., `network.sh`, `db_app.sh`, `scaling.sh`).
- **Verification (Requirement: Yes/No):** To ensure you meet the "Yes/No" measurement criteria, add a verification step at the end of each script that queries the resource to confirm it exists, for example:
 - `aws ec2 describe-vpcs --vpc-ids $VPC_ID`
- **Avoid Defaults:** The marking scheme specifically penalizes using generic or wrong defaults. Ensure your scripts explicitly define parameters like subnets, CIDRs, and security group rules.
- **Region Control:** Always set `--region us-east-1` in your CLI commands, as using the wrong region will result in a zero mark.

END TEST PROJECT

RULES & REGULATIONS

The competition will run to WorldSkills International Rules and Code of Ethics, Work Health and Safety is to Australian Standards and you can speak to your Australian Expert for further information. Refer to [Global Skills Website](#).

EXPERT CONTACT

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